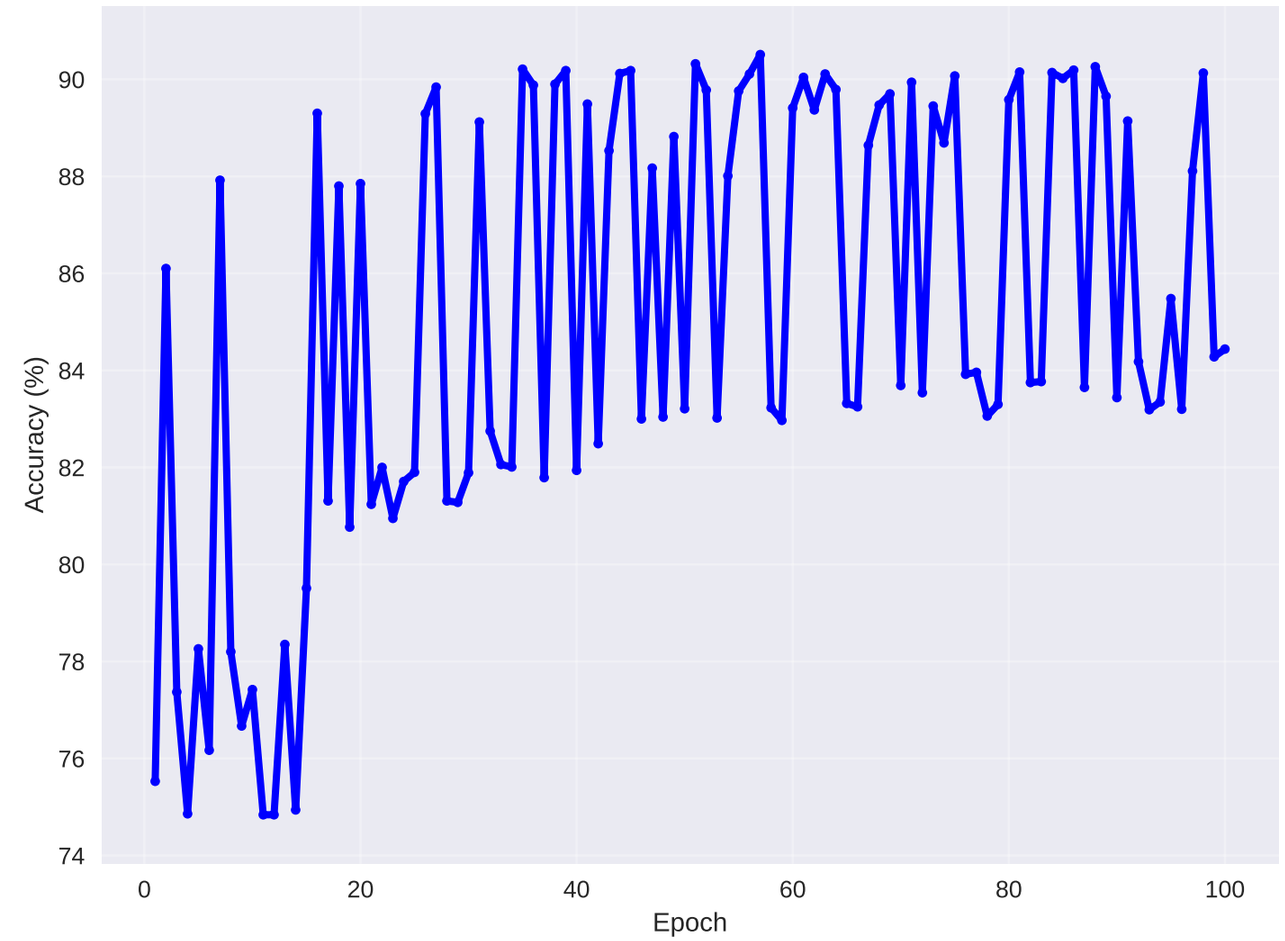
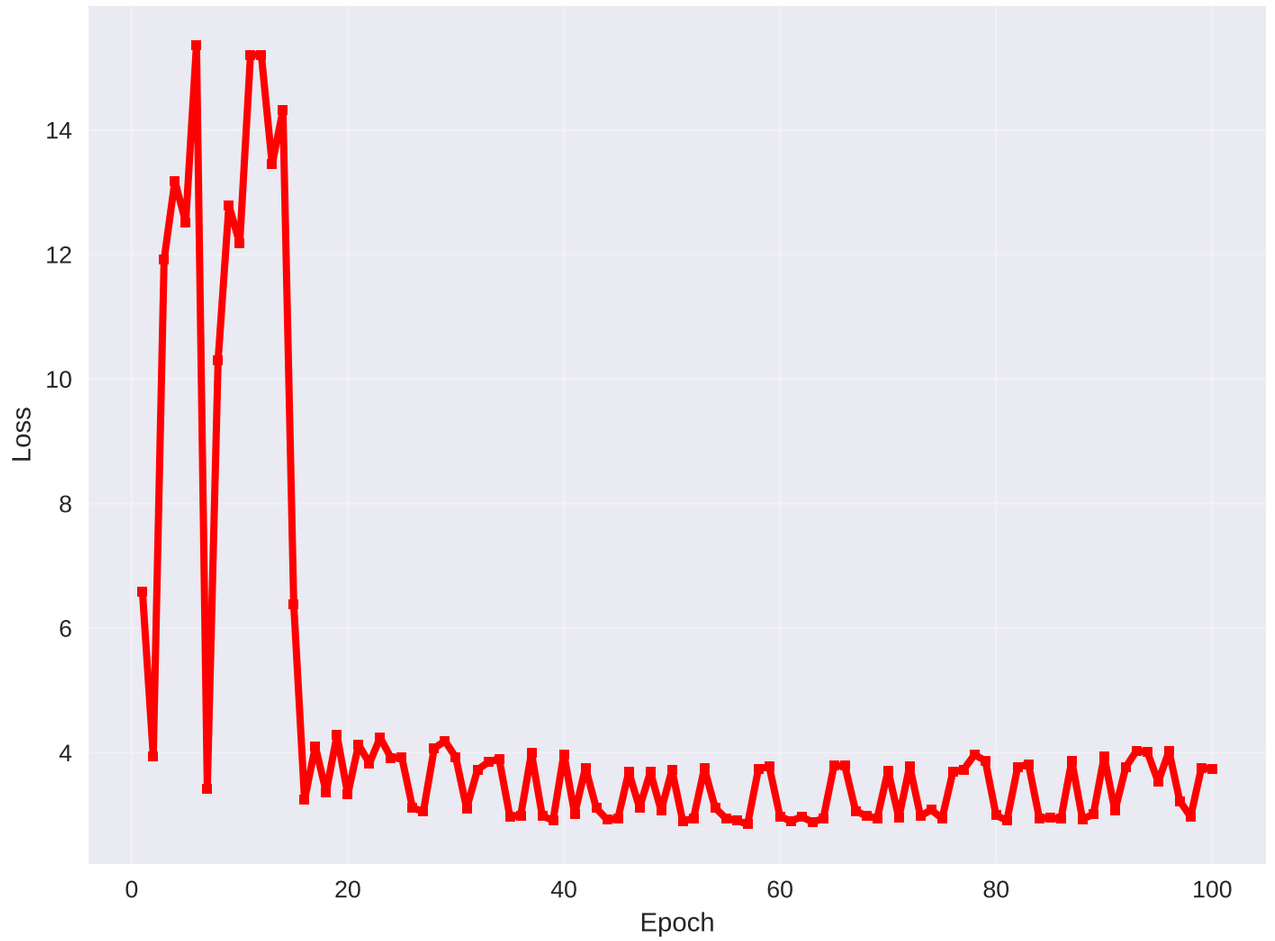
Federated Learning - Main Metrics Overview

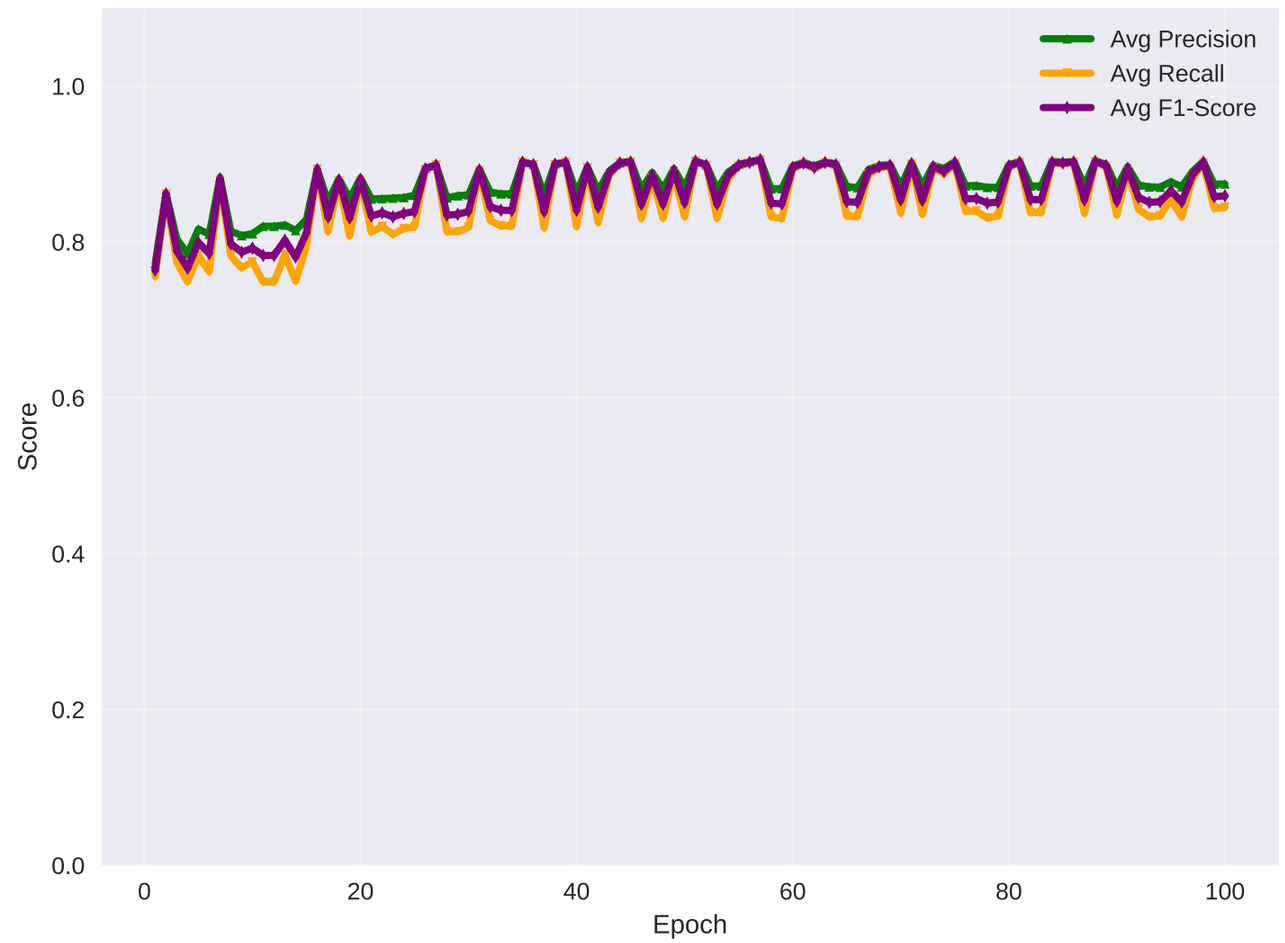
 Global Model Accuracy Evolution



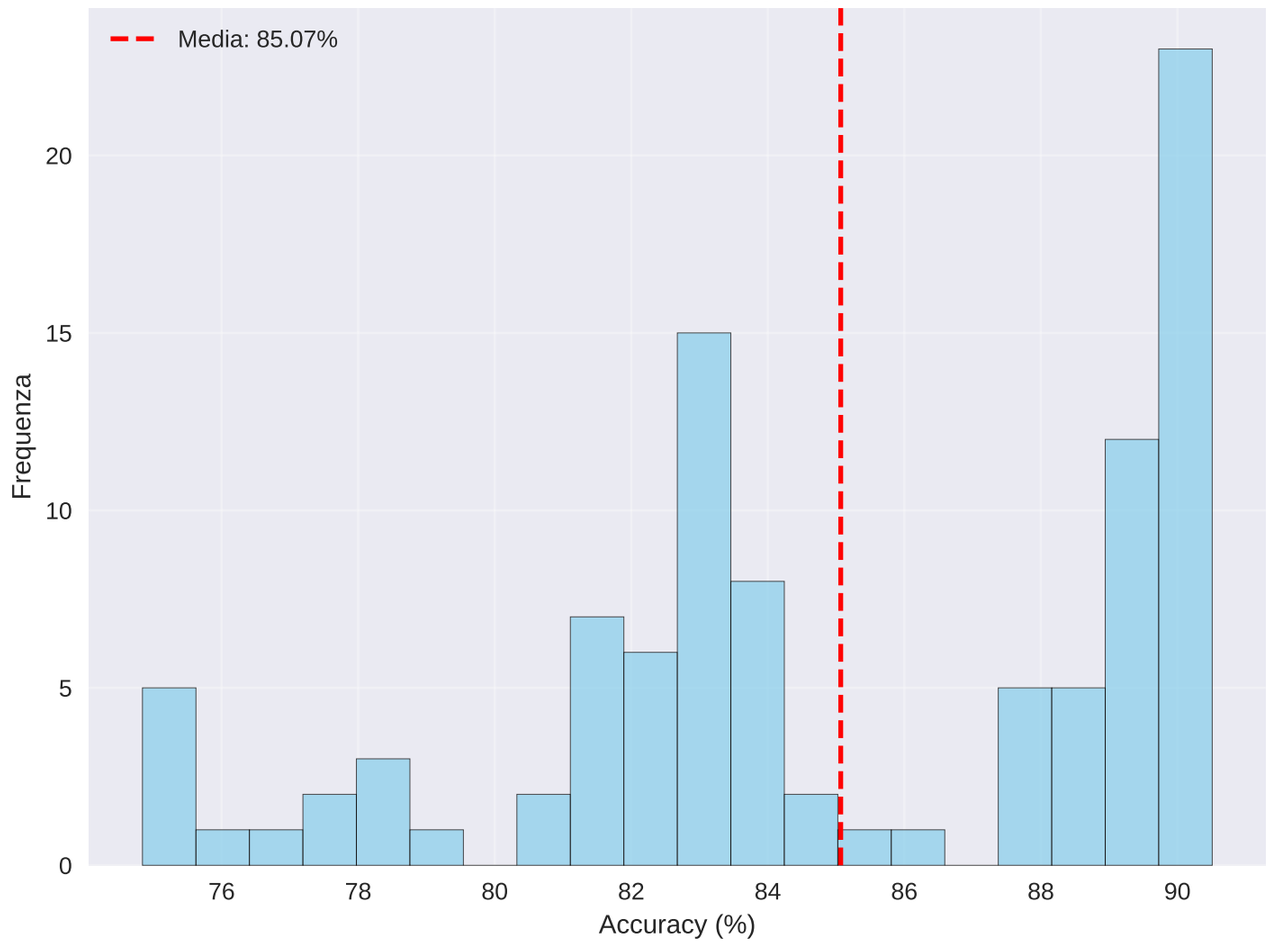
 Global Model Loss Evolution



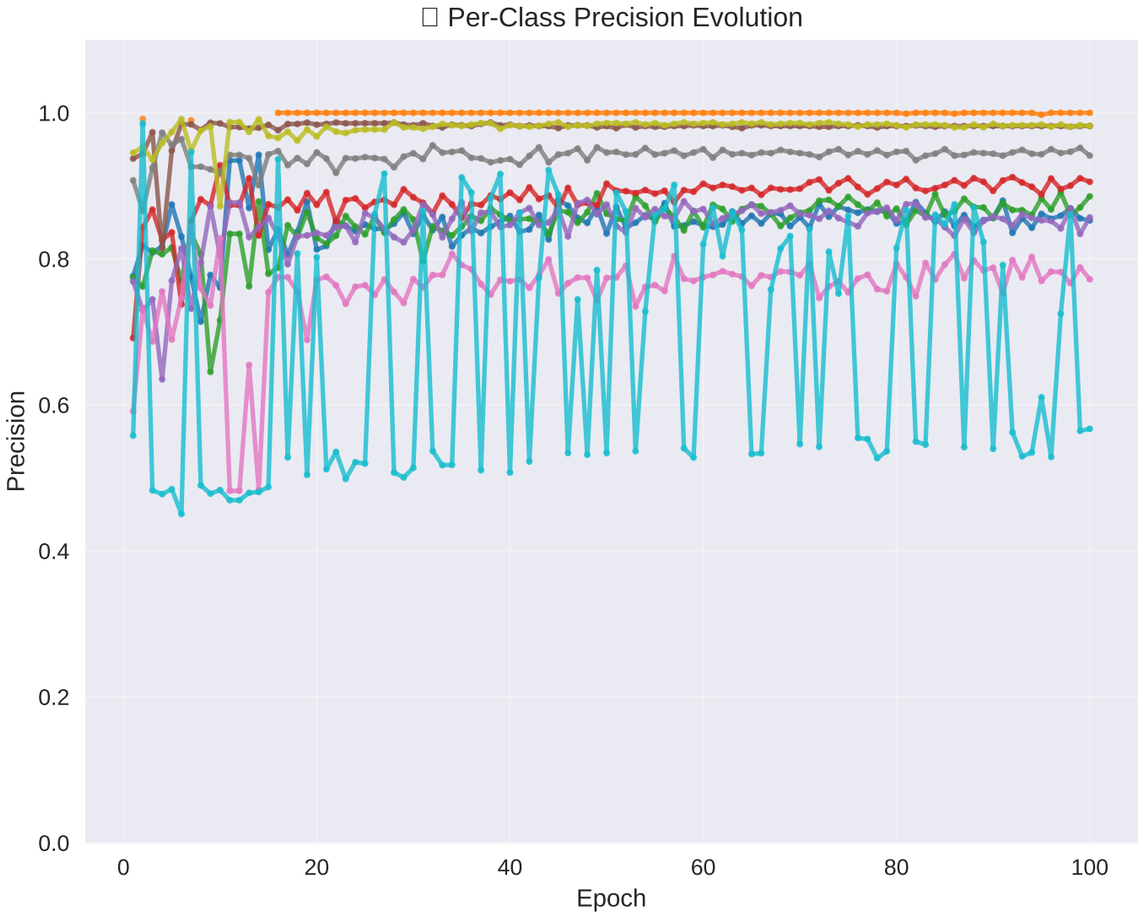
 Average Precision, Recall & F1-Score

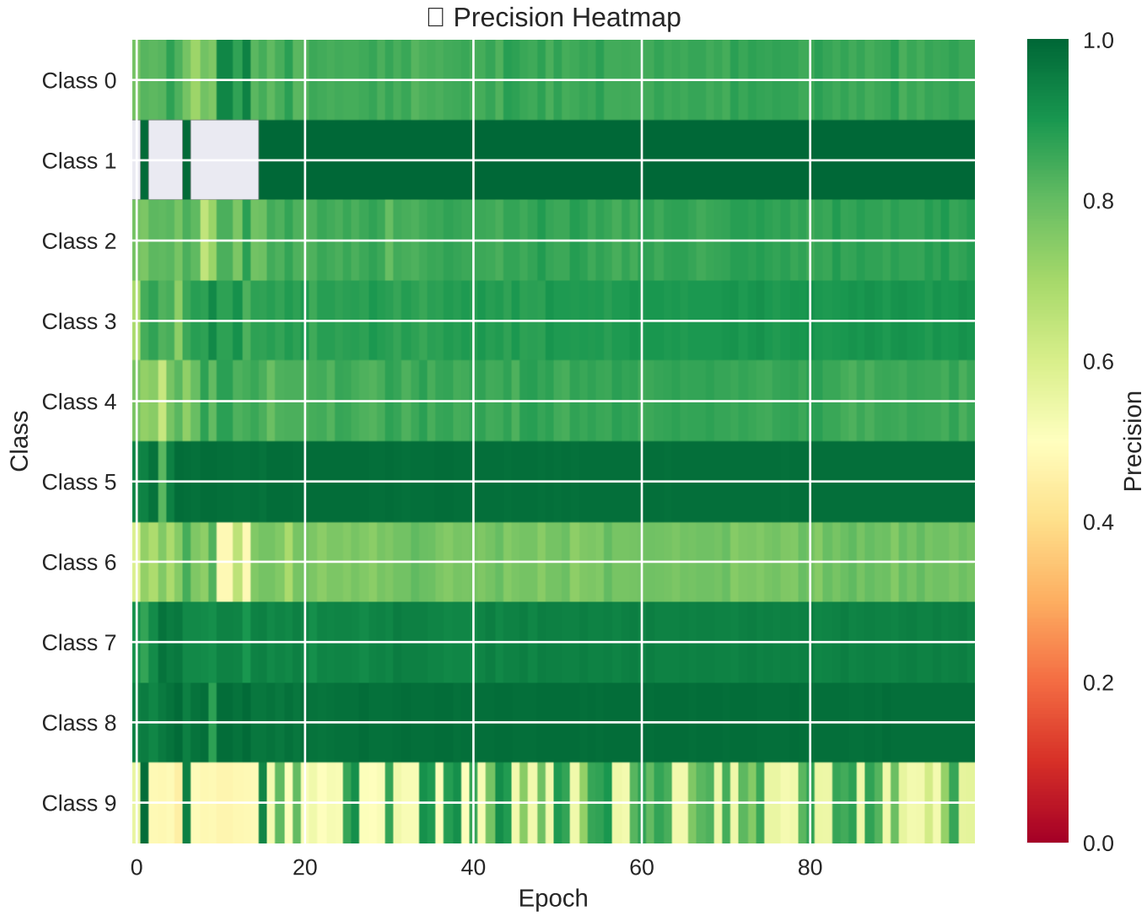


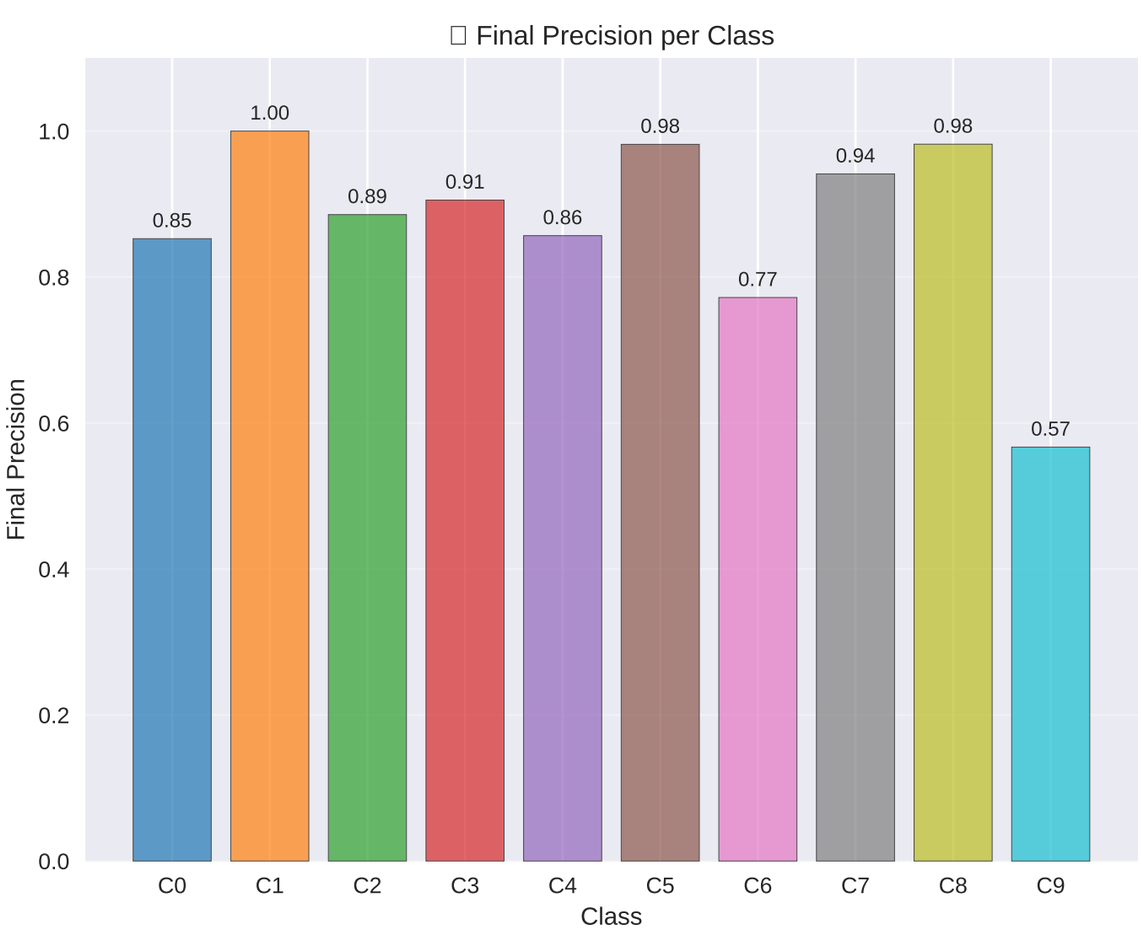
 Distribuzione Accuracy

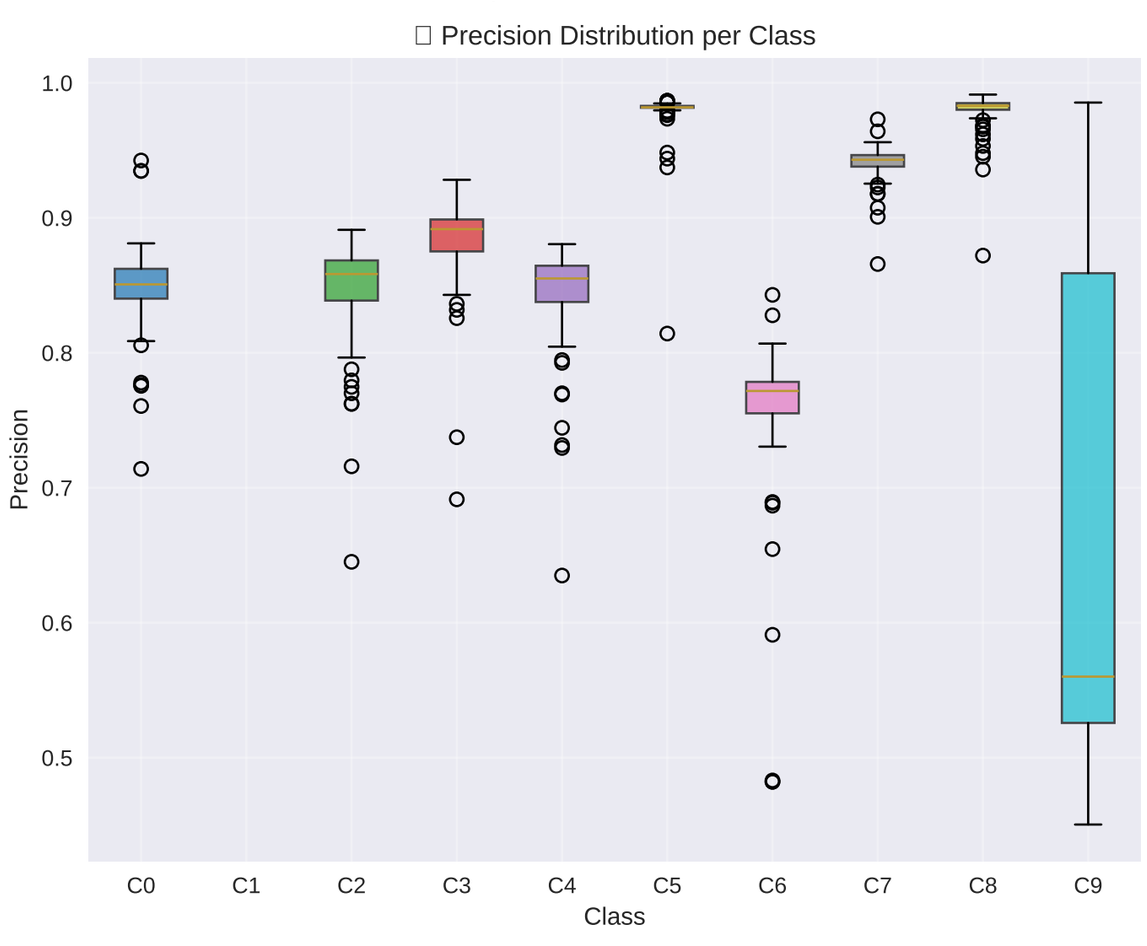


Per-Class Precision Analysis



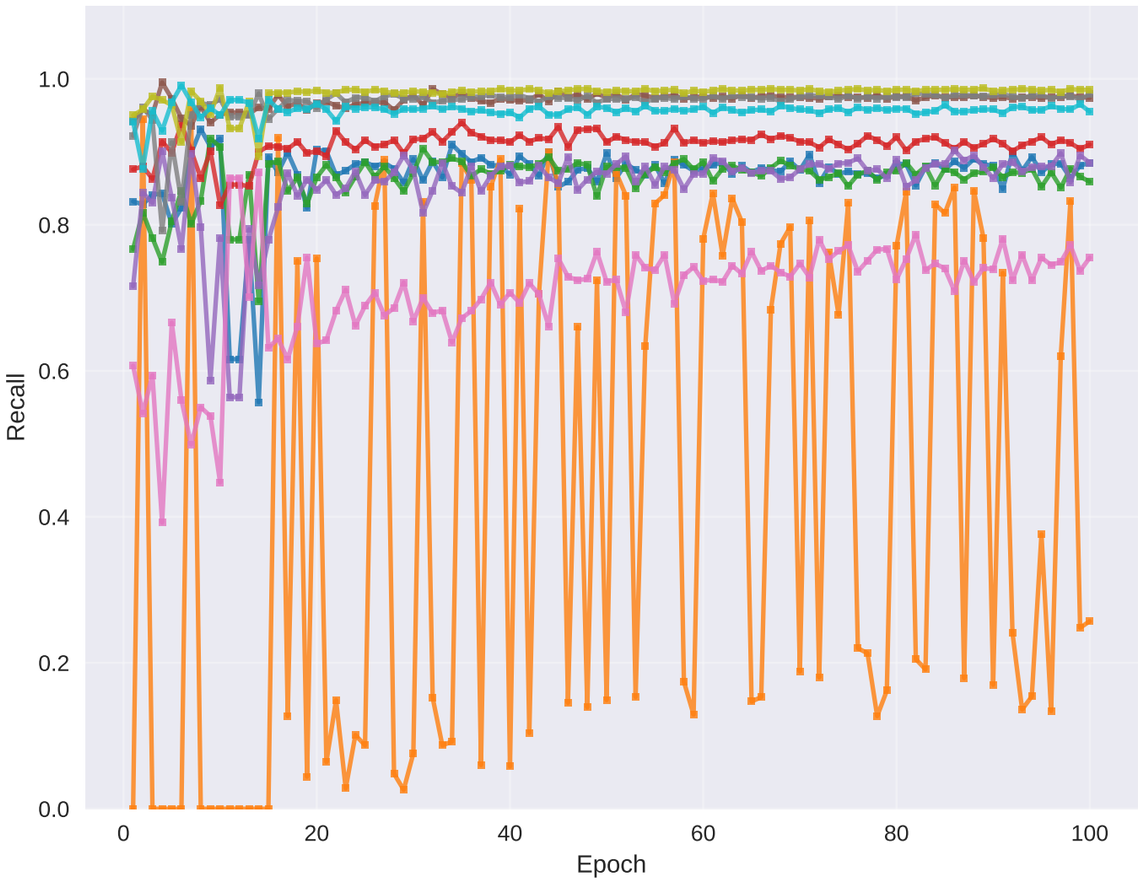




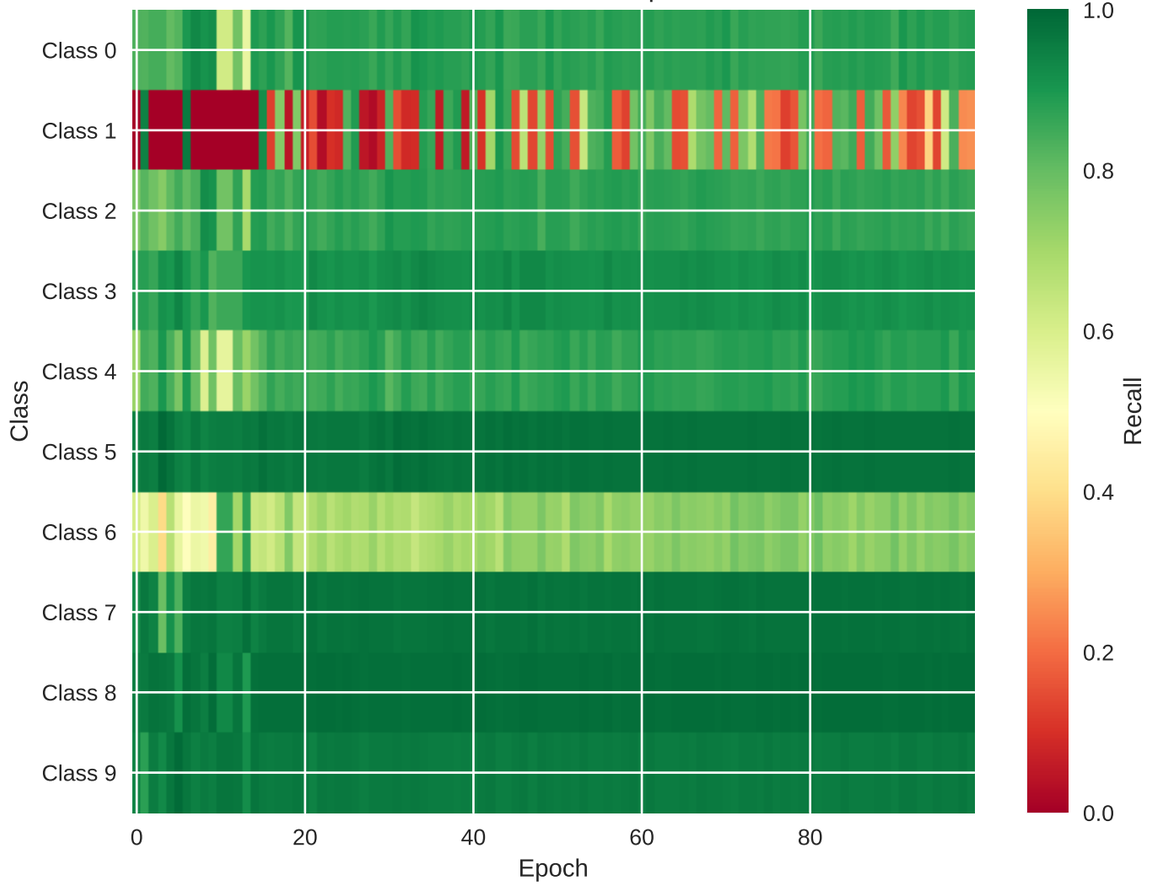


Per-Class Recall Analysis

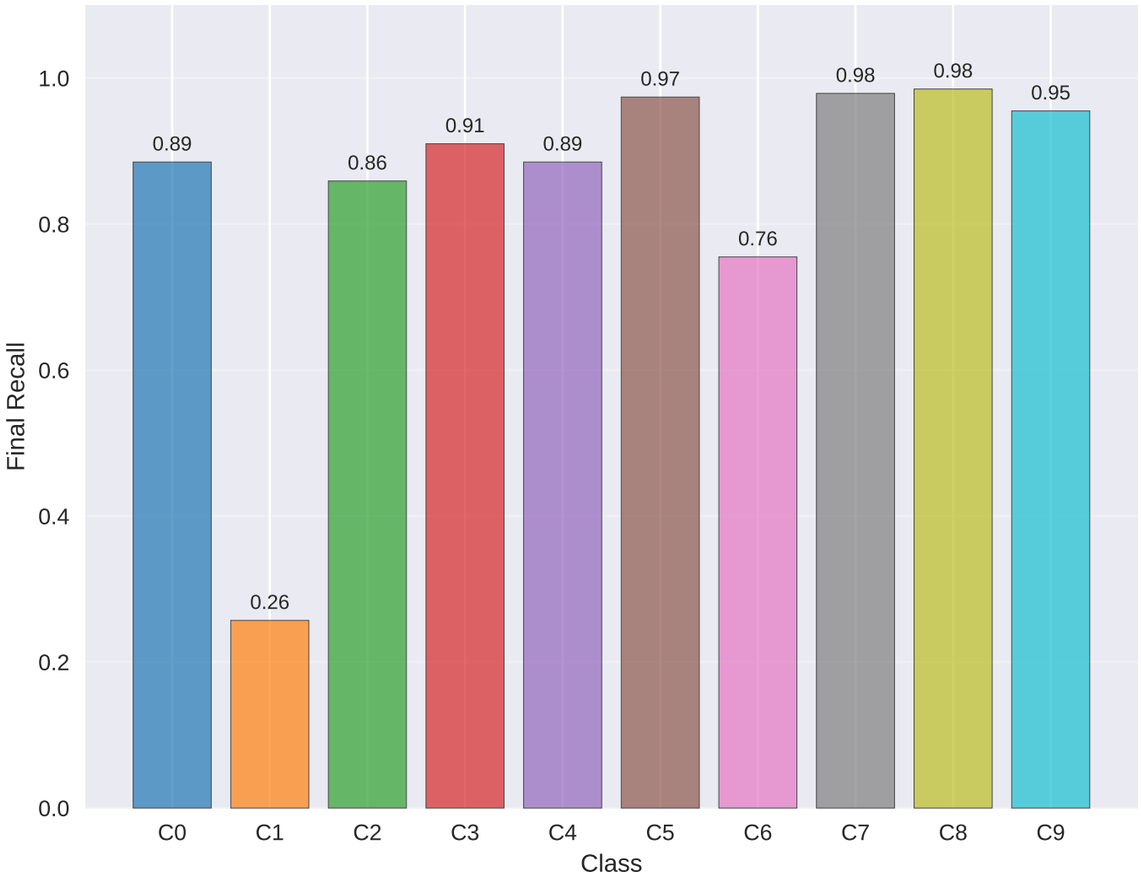
□ Per-Class Recall Evolution



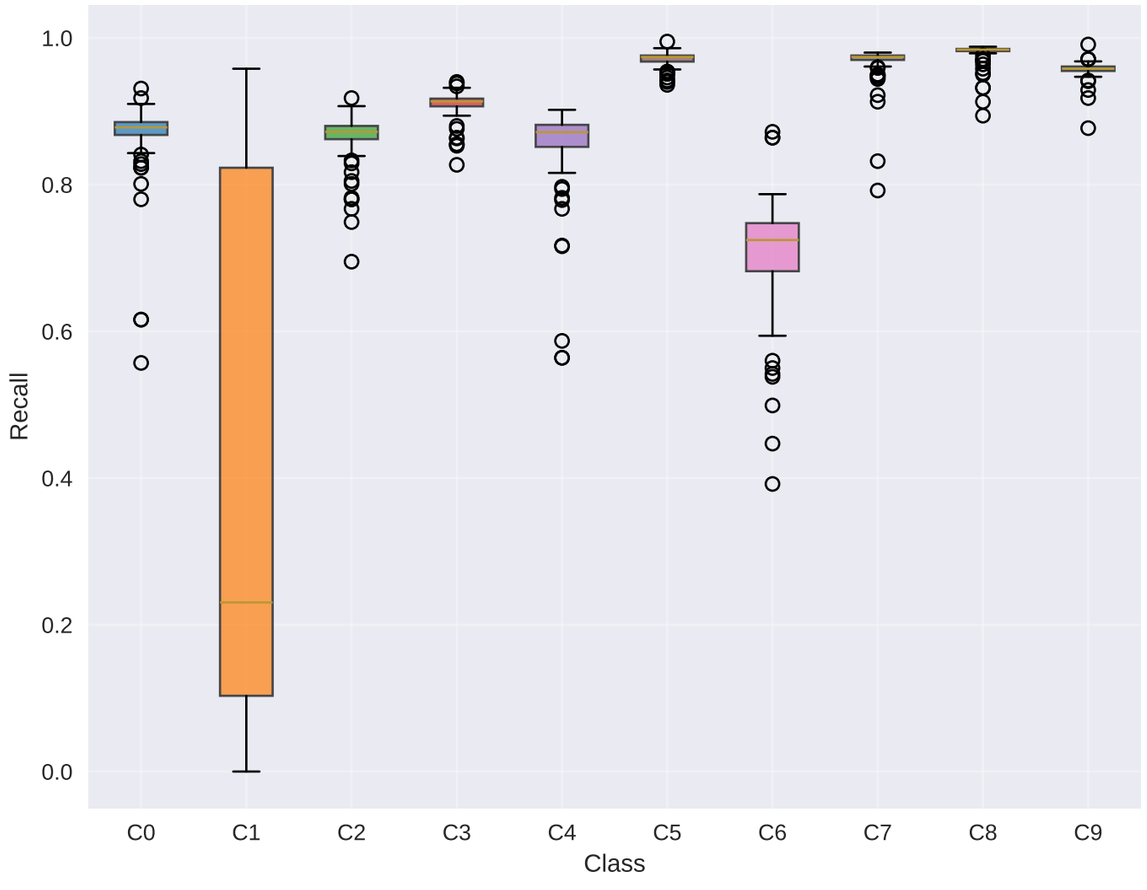
□ Recall Heatmap



□ Final Recall per Class

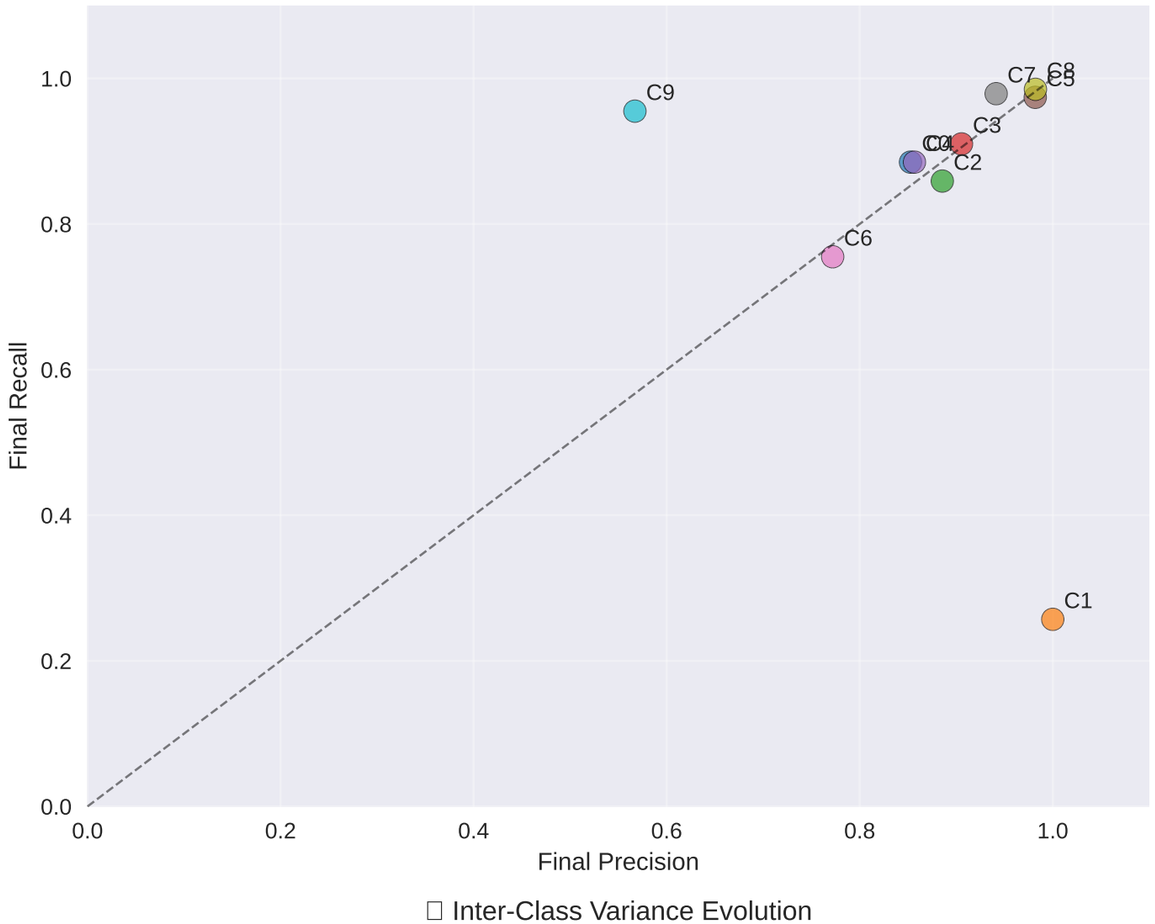


□ Recall Distribution per Class



Comparative Analysis

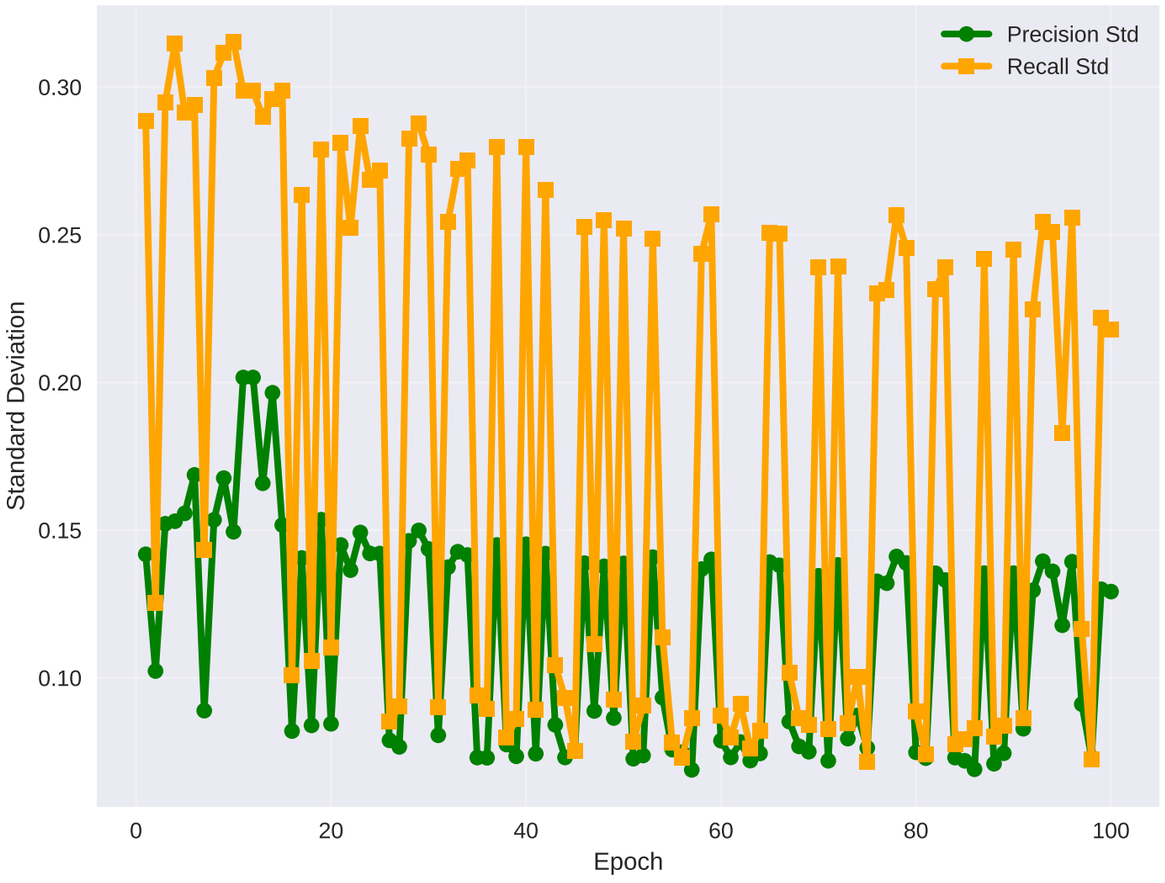
□ Precision vs Recall (Final Values)



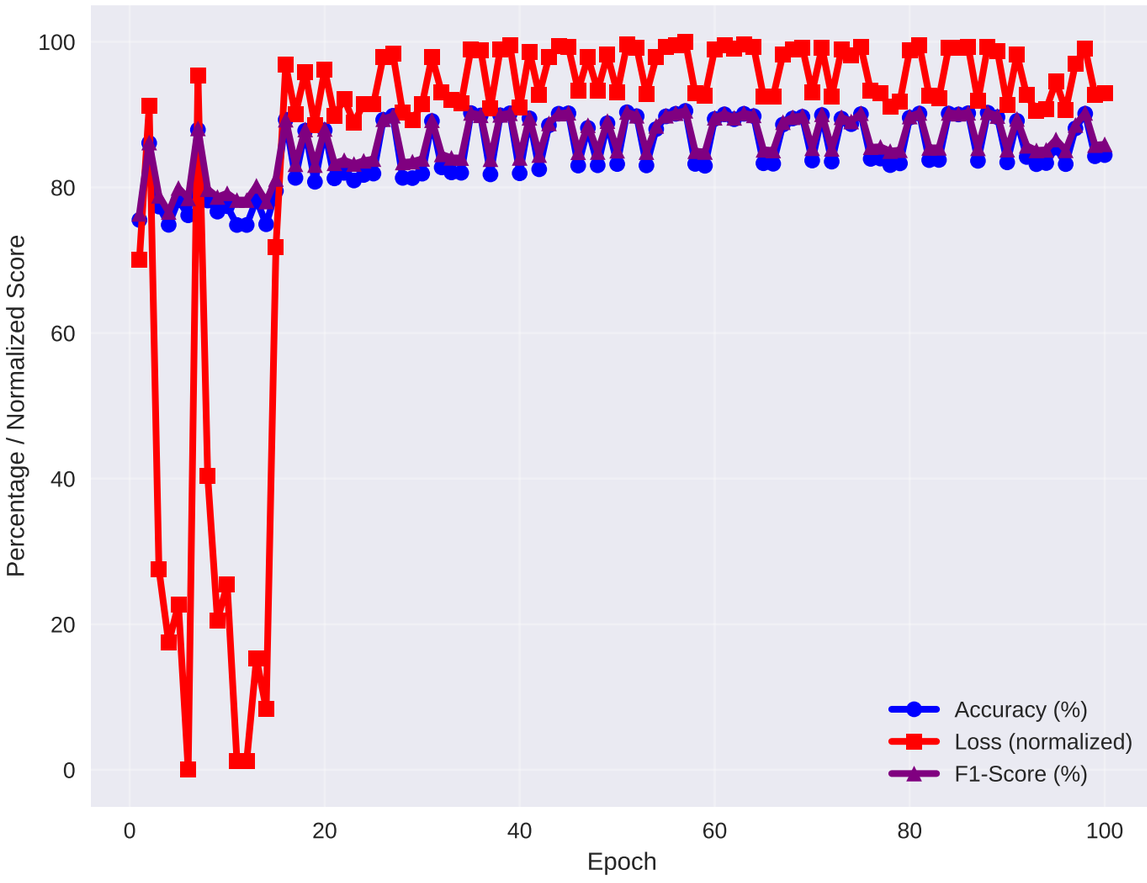
□ Per-Class F1-Score Evolution



□ Inter-Class Variance Evolution



□ Learning Curves Comparison



Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-05 15:36:18
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 84.44%
• Loss: 3.7365
• Average Precision: 0.874
• Average Recall: 0.844
• Average F1-Score: 0.859

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +8.91%
• Riduzione Loss: +2.8507
• Stabilità Accuracy (CV): 0.0541

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (1.000)
• Peggiore Precision: Classe 9 (0.567)
• Migliore Recall: Classe 8 (0.985)
• Peggiore Recall: Classe 1 (0.257)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.853, R=0.885, F1=0.868
• Classe 1: P=1.000, R=0.257, F1=0.409
• Classe 2: P=0.886, R=0.859, F1=0.872
• Classe 3: P=0.905, R=0.910, F1=0.908
• Classe 4: P=0.857, R=0.885, F1=0.871
• Classe 5: P=0.982, R=0.974, F1=0.978
• Classe 6: P=0.772, R=0.755, F1=0.763
• Classe 7: P=0.941, R=0.979, F1=0.960
• Classe 8: P=0.982, R=0.985, F1=0.984
• Classe 9: P=0.567, R=0.955, F1=0.712

VALUTAZIONE QUALITATIVA:
• Performance Globale: MEDIA
• Stabilità Training: INSTABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
• Implementare class balancing o weighted loss
• Migliorare stabilità con learning rate scheduling
```